

- 14** A and B same temperature (same pV).
So same r.m.s. speed.

C is incorrect:

$$\begin{aligned}c_{\text{r.m.s.}} &= \sqrt{\frac{3kT}{m}} \\ \Delta c_{\text{r.m.s.}} &= \sqrt{\frac{3k}{m}} (\sqrt{T_{\text{final}}} - \sqrt{T_{\text{initial}}}) \\ &\neq \sqrt{\frac{3k}{m}} (\sqrt{\Delta T}) \\ &\text{i.e. } \sqrt{\frac{3k}{m}} \left(\sqrt{\Delta \frac{pV}{Nk}} \right) \\ &\text{i.e. } \sqrt{\frac{3p}{Nm}} (\sqrt{\Delta V})\end{aligned}$$

15 thermometric properties change with temperature